
Homework 1

Music Era and Release-Market Classification

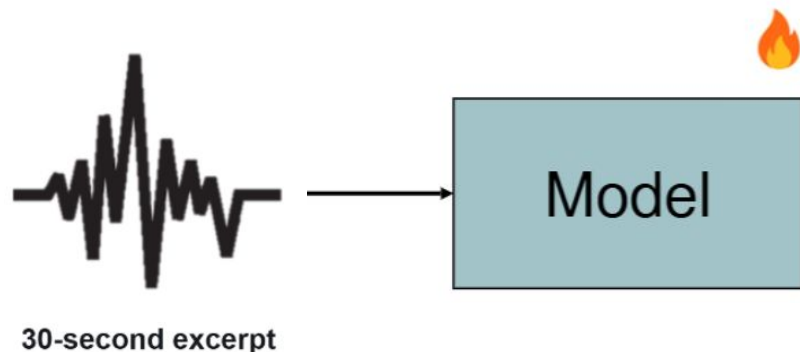
TA: 胡庭翊 (Ting-Yi Hu)

Office hour: Mon. 13:30~15:00 @BL505

Outline

- Overview
- Detailed Explanation
- Submission
- Scoring
- Rules
- Timeline

Overview: Single-label Classification



Six class scores

one label per excerpt

Top-1: highest score

Top-3: three highest scores

same prediction rule for A and B

Detailed Explanation

- Dataset
 - Task 1: Release-decade classification
 - Task 2: Release-market classification
- Suggested methods
- Evaluation
- Optional experiments

<https://drive.google.com/drive/folders/1C8RymiLbr-EGmkxh2Ap5TlybnJYqNsb4?usp=sharing>

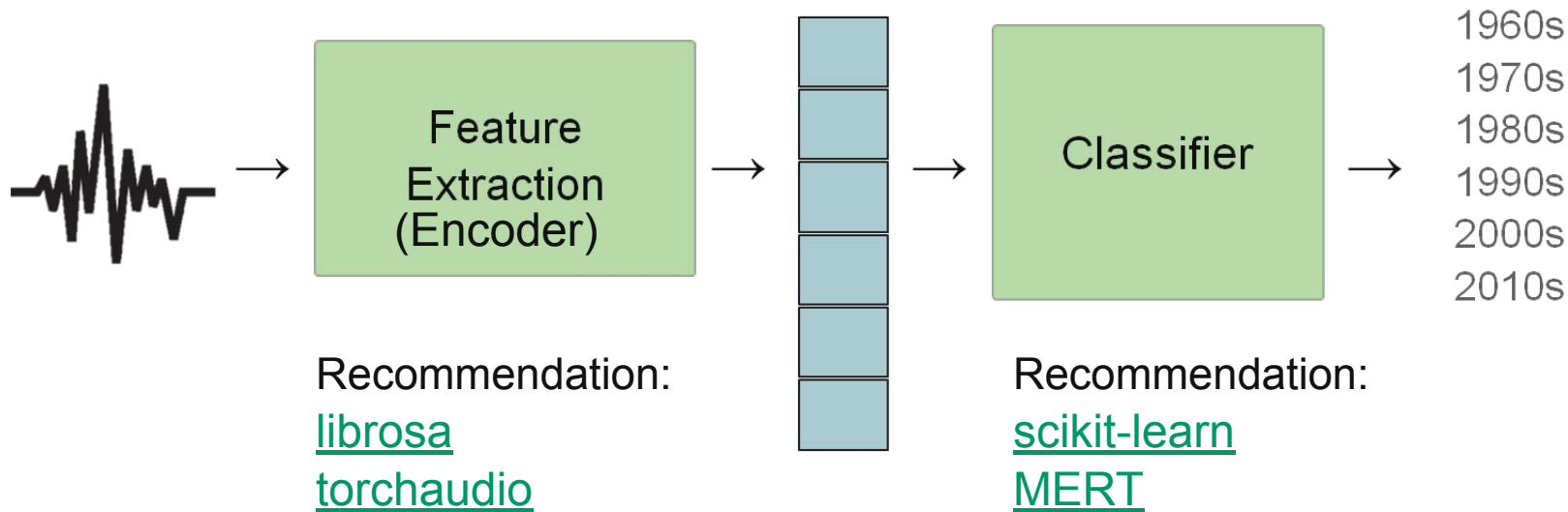
Dataset

- Source: Discogs-VI recordings with an available YouTube link
 - Dataset A: 1,290 recordings from six US release decades
 - Dataset B: 1,002 recordings from six release markets in the 1980s
- Each example is the middle 30 seconds of one recording
- Audio: WAV, mono, PCM signed 16-bit, 24 kHz
- Anonymous sample IDs and stripped title and artist metadata
- Artists do not overlap across train, validation, and test

```
dataset_A/  
train/  
  A_00001.wav  
validation/  
test/  
manifest.csv
```

```
dataset_B/  
train/  
validation/  
test/  
manifest.csv
```

Task 1: Release-Decade Classification



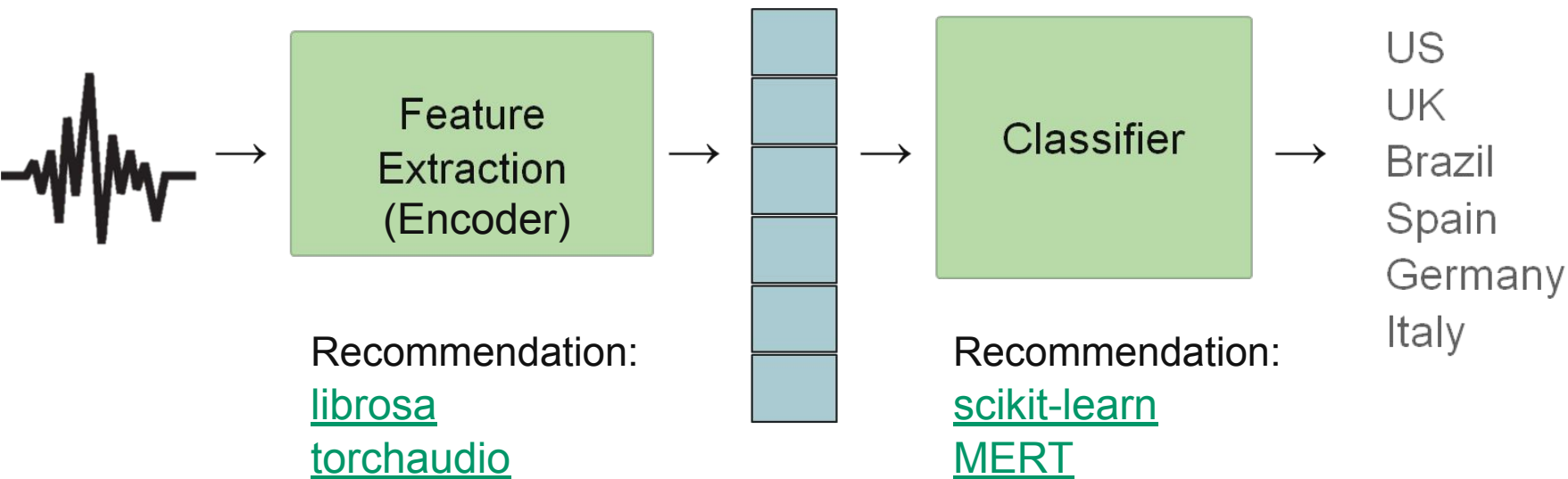
Dataset A: US releases, six balanced decade labels, 1,026 / 132 / 132 split

Task 1: What You Need to Do

- Train a model with any features extracted from the audio.
- Need to report **the features** you use and the **model implementation** clearly.
- Need to report the **validation** result with **confusion matrix, top1 accuracy, and top3 accuracy**.
- Remember to utilize standardization (e.g. mean, std), pooling and normalization to ensure consistent feature scales, reducing overfitting, and improving model stability and performance during training.
- Discuss whether errors occur mainly between neighboring decades

The report must explain what the model learned and where neighboring decades are confused.

Task 2: Release-Market Classification



Dataset B: 1980s releases, six balanced market labels, 798 / 102 / 102 split
Release market is not the artist's nationality, language, ethnicity, or recording location.

Task 2: What You Need to Do

- Train a model with any features extracted from the audio.
- Need to report **the features** you use and the **model implementation** clearly.
- Need to report the **validation** result with **confusion matrix, top1 accuracy, and top3 accuracy**.
- Remember to utilize standardization (e.g. mean, std), pooling and normalization to ensure consistent feature scales, reducing overfitting, and improving model stability and performance during training.
- Discuss whether errors occur mainly between release markets

Suggested Method: Short-Chunk CNN

- Convert the waveform into a log-mel spectrogram
- During training, sample a short segment from the 30-second excerpt
- Use a convolutional network to predict six class scores
- During evaluation, combine predictions from multiple segments

Audio → log-mel spectrogram → CNN → class probabilities

Suggested Method: Pretrained Audio Features

- Use MERT or another pretrained audio encoder
- Pool frame-level representations into one recording-level feature
- Train a classifier such as logistic regression, SVM, or MLP
- Compare frozen features with fine-tuning if resources allow

Audio → pretrained encoder → pooling → classifier

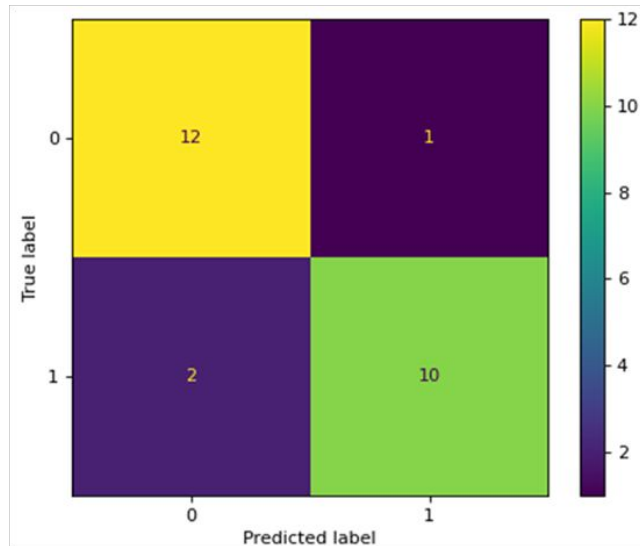
Evaluation

1. Confusion matrix

Compare predicted labels with the true labels. State whether the matrix shows counts or normalized values.

2. Top-k accuracy

Report Top-1 and Top-3. Top-3 counts a sample as correct when the true label appears among the three submitted labels.



Use train for fitting, validation for model selection, and test for final reporting.

Data Pre-processing / Data Augmentation

Audio processing

[librosa](#)

[torchaudio](#)

Source separation

[Open-Unmix](#)

[Spleeter](#)

[Demucs](#)

- Preserve cues that may carry decade or market information
- Avoid transformations that change the label meaning
- Compare the original mixture with vocal and accompaniment stems
- Report every preprocessing and augmentation step

Optional Experiments

- Compare 5, 10, 15, and 30 seconds of audio
- Use multiple excerpts from the same recording during training
- Treat Task 1 as year regression or hierarchical prediction
- Compare mixture, vocal, and accompaniment inputs for Task 2
- Visualize learned embeddings with [t-SNE](#) or [UMAP](#)
- Analyze confusion patterns and dataset bias

Audio Language Model:

- Run an audio language model on both Dataset A and Dataset B
- Use every sample in the selected split and return one valid answer for every sample
- Describe the prompt and output parsing method
- Report Top-1, Top-3, confusion matrices, and invalid-output handling
- Compare at least two prompt designs or explain why you selected one prompt

[Qwen2-Audio](#)

[Audio Flamingo](#)

Submission

- Report PDF to NTU COOL
- Prediction JSON to NTU COOL
- Source code and trained checkpoint in an open-access cloud-drive folder
- requirements.txt and README.md in the same folder

We may select submissions and run inference. The uploaded files must reproduce the submitted predictions.

Do not upload the released dataset, model caches, or unrelated checkpoints.

Report

- Use a 16:9 PPT or PPT-like format and submit it as PDF
- File name: <studentID>_report.pdf
- The report must be understandable without oral explanation
- Explain methods and results for both tasks
- Include at least one confusion matrix for each task
- Discuss errors, limitations, and reproducibility
- Cite every public codebase, pretrained model, and paper used

About 10 pages is a suggested length, not a strict limit.

Prediction

- File name: <studentID>.json
- Include every Dataset A and Dataset B test sample exactly once
- Return **top-3 labels in descending confidence order**

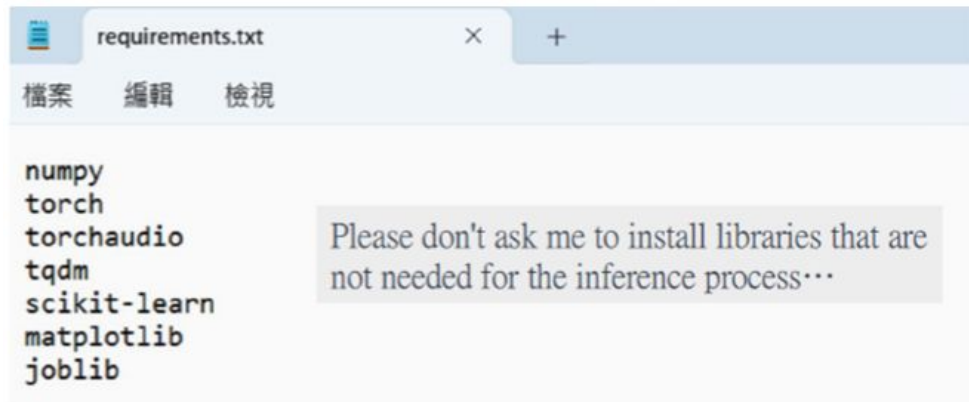
```
{
  "dataset_A": {
    "A_0072adbc81aa": ["2010s", "2000s",
"1990s"]
  },
  "dataset_B": {
    "B_00c58205be82": ["UK", "US",
"Germany"]
  }
}
```

Code & model checkpoint

- Upload all your source code and model to a cloud drive, open access permissions, and then upload the link to the NTU Cool assignment HW1_report in comments, as well as include it on the first page of the report
- You will need to upload requirements.txt
- I'll run :

```
pip install -r requirements.txt
```

If you have used third-party programs that cannot be installed directly via 'pip install,' please write the URL and install method command by command on **your readme file**.



The screenshot shows a code editor window titled 'requirements.txt' with tabs for '檔案' (File), '編輯' (Edit), and '檢視' (View). The file content lists the following dependencies: numpy, torch, torchaudio, tqdm, scikit-learn, matplotlib, and joblib. A light gray warning box on the right side of the editor contains the text: 'Please don't ask me to install libraries that are not needed for the inference process...'

```
numpy  
torch  
torchaudio  
tqdm  
scikit-learn  
matplotlib  
joblib
```

Please don't ask me to install libraries that are not needed for the inference process...

Code

- You will also need to upload readme file to guide me on how to perform inference on your model. (I'll use the same test set, ensure I can run your code with the test set path in my device.)
- The inference code should output the top 3 predictions in descend accuracy order

Scoring

Report: 50%

Accuracy: 50%

For each task:

$$S = (\text{Top-1} + 0.5 \times \text{Top-3})$$

$$\text{Accuracy points} = 50 \times (S_A + S_B) / 2$$

A-: All basic requirements are completed properly.

A: The work goes beyond the basic requirements in implementation, performance, or analysis.

A+: A creative approach with exceptionally strong implementation and accuracy may receive an A+.

Class Sharing: Three students with representative submissions will be invited to present their work in class.

Rules

- Do not cheat
- Do not use additional labeled audio unless the course explicitly permits it
- Do not train or tune on validation or test examples
- Do not hard-code predictions from the released labels
- You may use public code and pretrained models with proper citation
- Your uploaded code and checkpoint must reproduce the submitted predictions

Timeline

- W2 – 09/15 (Tuesday): Announcement of HW1
- W5 – 10/05 (Monday 23:59pm): Deadline

Late submission: 1 day (-20%), 2 days (-40%), 3 days (-60%), after that (no score)

ALL things you need to do before 10/05 23:59

- StudentID_report.pdf
- Cloud drive link
 - readme file
 - Requirements.txt
 - Codes and model to run inference
 - Others codes
- Test set prediction

